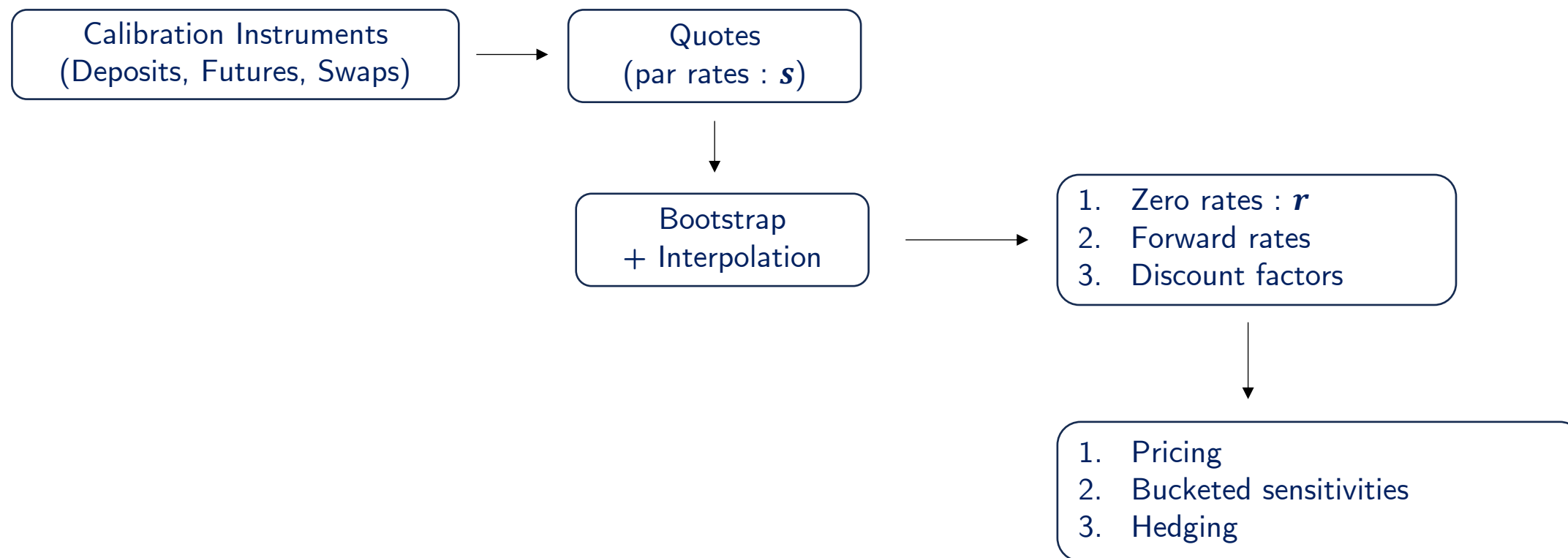


RFR Regime

(7. Jacobian for Fast curve building)

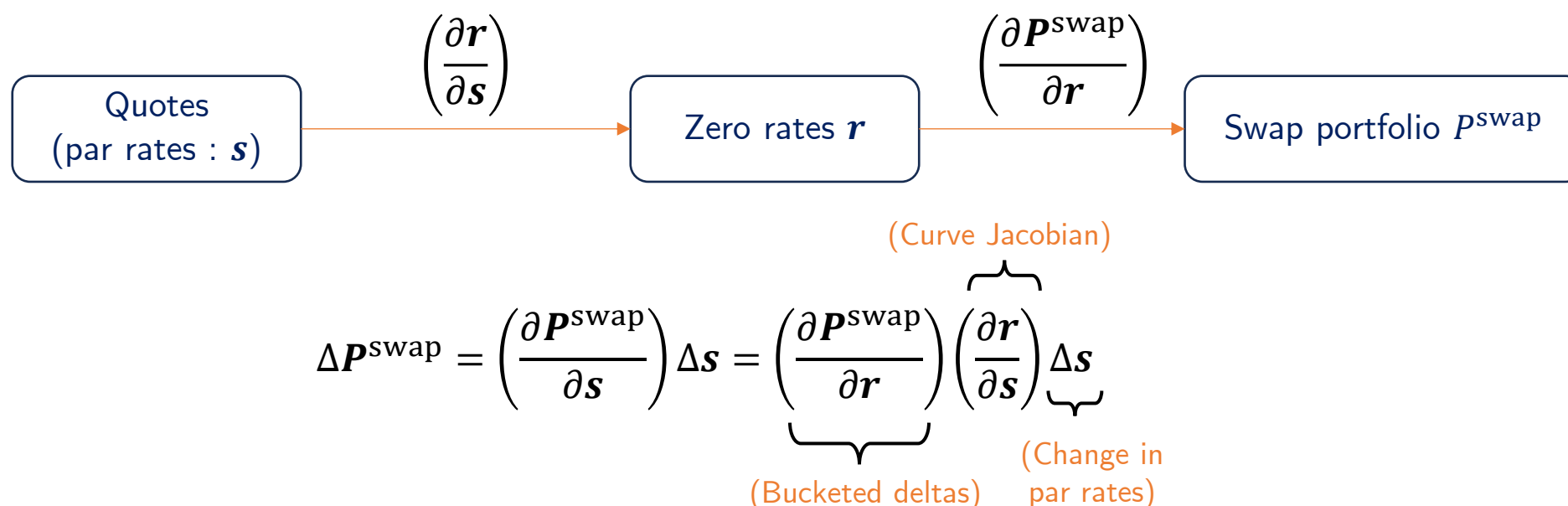
Rate Curve Flowchart

- At a high-level, a rate desk performs the following steps (consider only linear derivatives for simplicity)



The Problem Statement

- ▶ Zero curve is the base curve against which **pricing functions** and **bucketed sensitivities** (delta, gamma etc.) are calculated and stored. Market quotes (par rates) change frequently which require us to **recalibrate** our zero curves in **real time** and then reprice the instruments.
- ▶ We do not need to recalibrate again and again as we can pre-compute and store the Jacobian. This will help rebuild fast zero curves and reprice portfolio



Computing the Curve Jacobian

- ▶ The Curve Jacobian can be computed analytically

For swap rates, the Jacobian can be populated as follows

$$\begin{matrix} & k \rightarrow \\ j \downarrow & \begin{pmatrix} \dots & \dots & \dots \\ \dots & \dots & \dots \\ \dots & \dots & \dots \end{pmatrix} \end{matrix} \quad \left(\frac{\partial S_j}{\partial r_k} \right) = \frac{S_j v_k T_k \mathbf{1}_{k \leq j} + v_j T_j \mathbf{1}_{j=k}}{\sum_{i=1}^j v_i}$$

where, j is the maturity

s_j : swap rate , r_j : zero rate, v_j : discount factor

Example

	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
1																			
2																			
3																			
4																			
5																			
6																			
7																			
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27																			
28																			

$$\left(\frac{\partial S}{\partial r}\right)$$

$$T_j \quad s_j \quad v_j$$

$$j \downarrow$$

	1	2	3	4	5
1	0.9661	0.93811	0.92219	0.89209	0.85804
2	0.9661	0.93811	0.92219	0.89209	0.85804
3	0.9661	0.93811	0.92219	0.89209	0.85804
4	0.9661	0.93811	0.92219	0.89209	0.85804
5	0.9661	0.93811	0.92219	0.89209	0.85804

$$k \rightarrow$$

	1	2	3	4	5
1	1.04	0.00	0.00	0.00	0.00
2	0.02	1.02	0.00	0.00	0.00
3	0.01	0.02	1.01	0.00	0.00
4	0.01	0.01	0.02	0.99	0.00
5	0.01	0.01	0.02	0.02	0.97

$$T_k$$

$$v_k$$

$$\left(\frac{\partial S_j}{\partial r_k}\right) = \frac{S_j v_k T_k \mathbf{1}_{k \leq j} + v_j T_j \mathbf{1}_{j=k}}{\sum_{i=1}^j v_i}$$

Curve Jacobian

0.9661	0	0	0	0
-0.0157	0.98297	0	0	0
-0.0088	-0.0179	0.99426	0	0
-0.007	-0.0142	-0.0217	1.01269	0
-0.006	-0.0122	-0.0187	-0.0253	1.03465

New Curve

$$r_{new} = r_{old} + \left(\frac{\partial r}{\partial s}\right) \Delta s$$

Year	r
1	3.458%
2	3.204%
3	2.710%
4	2.864%
5	3.072%

Δs
0.01%
0.01%
0.01%
0.01%
0.01%

Swap Jacobian

$$\left(\frac{\partial P}{\partial s}\right) = \left(\frac{\partial P}{\partial r}\right) \left(\frac{\partial r}{\partial s}\right)$$

-96.60	0.00	0.00	0.00	0.00
-0.05	-190.49	0.00	0.00	0.00
0.00	0.01	-282.59	0.00	0.00
0.04	0.09	0.14	-371.59	0.00
0.04	0.09	0.13	0.18	-457.31

Swap PnL

$$\Delta P = \left(\frac{\partial P}{\partial s}\right) \Delta s$$

Maturity	PnL
1Y	\$ (96.60)
2Y	\$ (190.54)
3Y	\$ (282.58)
4Y	\$ (371.31)
5Y	\$ (456.86)

Total PnL \$ (1,397.89)